



Electric field generated by axial longitudinal vibration modes of microtubule

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ABSTRACT

Microtubules are electrically polar structures fulfilling prerequisites for generation of oscillatory electric field in the kHz to GHz region. Energy supply for excitation of elasto-electrical vibrations in microtubules may be provided from GTP-hydrolysis; motor protein–microtubule interactions; and energy efflux from mitochondria. We calculated electric field generated by axial longitudinal vibration modes of microtubules for random, and coherent excitation. In case of coherent excitation of vibrations, the electric field intensity is highest at the end of microtubule. The dielectrophoretic force exerted by electric field on the surrounding molecules will influence the kinetics of microtubule polymerization via change in the probability of the transport of charge and mass particles. The electric field generated by vibrations of electrically polar cellular structures is expected to play an important role in biological self-organization.

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1. Introduction

Cytoskeleton, which consists of microtubules, actin filaments, and intermediate filaments, is considered to be an organizing structure of the eukaryotic cell (Alberts et al., 2008). Microtubules (MTs) are of special interest since they manifest several features that distinguish them from other subcomponents of the cytoskeleton. MTs, which consist usually of 13 protofilaments (PT), resemble hollow tubes with inner and outer diameter of 17 nm and 25 nm, respectively (Fig. 1a). The building subunits of MT are the tubulin heterodimers composed of α - and β -tubulin (Fig. 2). The heterodimers have high electric dipole moment of over 1000 Debye (10^{-26} C m) (Tuszyński et al., 2002; Merzhin et al., 2004); they are responsible for the high electrical polarity of MTs. In the interphase of the cell cycle, microtubules are radially organized with their chemical minus ends embedded in a centrosome (organizing center), which is located in the center of the cell near the cell nucleus. There are several hundreds of MTs in a cell, depending on the organism and the cell type. In the interphase microtubules undergo dynamic growth (polymerization) and shrinkage (depolymerization), so-called “dynamic instability” (Fig. 1b). During the existence of mitotic spindle, MTs are subject to treadmilling, i.e. they polymerize at the plus end and depolymerize at the minus end (chemical plus and minus).

1.1. Microtubule Vibrations

Elastic and vibrational properties of MTs have been studied by several authors. Sirenko et al. (1996) analyzed vibrations of microtubules in a fluid modeled as elastic cylindrical shell with isotropic mechanical properties. Maximum frequency of tens of GHz corresponded to nonradiative elastic waves localized at the MT wall.

Pokorný et al. (1997) analyzed the vibrations of MTs modeled as one dimensional chain of mass particles with translation symmetry. Vibrations of cellular structures have been considered to be viscously overdamped in cytosol and thus not realistic (Foster and Baish, 2000; Adair, 2002). However, most of the water in the cell has different properties than bulk water (Preparata, 1995; Del Giudice et al., 2000, 2005; Zhadin and Giuliani, 2006; Zhadin et al., 2007) due to the molecular crowding and consequent large hydration surfaces. If the slip layer condition for MT longitudinal vibrations is taken into account, vibrations may be excited (Pokorný, 2004, 2003, 2005).

Portet et al. (2005) describe the microscopic dynamical properties of MTs using a discrete model based on a lattice of dimers. It was pointed out that various experimental methods yielded wide range of values of Young's and shear modulus between 1 Pa (Janmey et al., 1991; Sato et al., 1988) up to 1 GPa (Portet et al., 2005). This would be explainable by the fact that mechanical properties of microtubules are anisotropic and orthotropic.¹ This was showed by Kis et al. (2002) who obtained the most accurate values for both the Young's ($E > 100$ MPa) and shear modulus ($G = 1.4 \pm 0.4$ MPa).

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¹ Orthotropic–anisotropic in the directions perpendicular to each other.

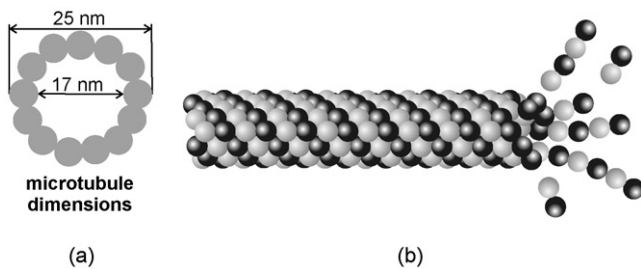


Fig. 1. (a) Cross section of microtubule. (b) Detail of chemical plus end of a microtubule undergoing dynamic instability.

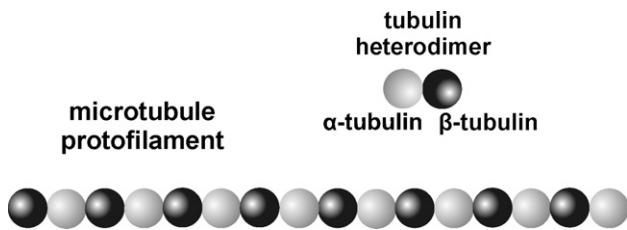


Fig. 2. Structural subunits of microtubules: protofilament composed of tubulin heterodimers.

The oscillation modes involving a change in the diameter of the MTs strongly depend on the shear modulus (Kasas et al., 2004).

Tuszyński et al. (2005) reviewed characteristic elastic properties of microtubules. Wang et al. (2006) and Wang and Zhang (2008) developed orthotropic elastic shell model to study microtubule vibrations and determined more realistic frequencies of radial and circumferential modes than that provided by Sirenko et al. (1996) which was confirmed by Qian et al. (2007).

It can be seen from above given references that frequencies of MT vibration modes may span over broad spectral range from kHz to GHz. Due to non-linearities and to mechanical anisotropy of MTs, frequency mixing and energy transfer between the modes, one may expect various frequencies of oscillations.

1.2. Energy Supply to Microtubules

Excitation of mechanical vibrations in MTs, which are electrically polar, would give rise to electrical oscillations. To excite and sustain vibrations the energy needs to be supplied. In the case when at least a part of energy supplied to MTs excites elastic vibrations of electrically polar structures, generated oscillatory electric field has the same frequency as that of elastic MT vibrations. We analyzed three possible sources of energy supply to MTs (Jelínek et al., 2009):

- Energy released from hydrolysis of guanosine triphosphate (GTP) in processes of dynamic instability of MTs (Fig. 2) is the energy supply dependent on internal MT process. β -Tubulin bound GTP is hydrolyzed to guanosine diphosphate (GDP) after polymerization of tubulin heterodimer. In the interphase the half lifetime of individual unstable MTs (subject to dynamic instability) is about 5–10 min (Alberts et al., 2008). Energy from hydrolysis of GTP stored in MT lattice is estimated to be 1.7 kcal/mol^2 (Caplow et al., 1994; Caplow, 1995; Caplow and Shanks, 1996). The corresponding power input to MT lattice due to GTP hydrolysis is about $1.8 \times 10^{-14} \text{ W/mm}$ of MT (if we accept 5 min as a half life of a single MT) which range roughly corresponds to the previous estimate (Pokorný, 1999; Pokorný and Wu, 1998). Assuming 100 μm total length of MT structure in a cell (the length is assumed for

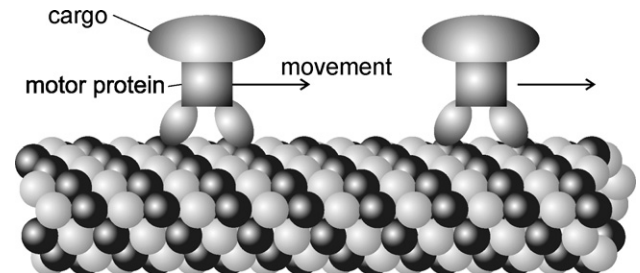


Fig. 3. Movement of the motor proteins on the microtubule, cargo not in scale.

yeast cell (King and Hyams, 1982; Winey et al., 1995; Sobel, 1997)), one obtains the power of about $3 \times 10^{-18} \text{ W}$ in whole microtubular network. In the M phase, the dynamics of polymerization and depolymerization is increased about 20-fold, for instance by treadmilling in the metaphase.

- Energy transferred from moving of motor proteins and their interaction with MTs may also contribute to excitation of vibrations in MTs (Fig. 3). Motor protein dynein moves towards the cell center (minus end of a MT) and kinesin moves towards the cell periphery (plus end of a MT). One ATP (adenosine triphosphate) molecule is hydrolyzed for each step of motor protein. One step of motor protein is ca. 8 nm long, corresponding to axial dimensions of tubulin heterodimer. Energy released during ATP hydrolysis under cellular ATP concentrations (1–10 mM) is ca. 14 kcal/mol^3 which corresponds to activation energy of motor proteins (Kawaguchi and Ishiwata, 2001, 2000; Pelling et al., 2004). Motor proteins move with a velocity ca. $0.1\text{--}2 \mu\text{m/s}$ (Alberts et al., 2008). Assuming four motor proteins on each of 100 MTs in the cell, each motor protein moving with average velocity of 800 nm/s, power exerted by all motor proteins in a cell is about $4 \times 10^{-15} \text{ W}$. Assuming that 1–25% of this power excites vibrations in MTs, it corresponds to $4 \times 10^{-17}\text{--}10^{-15} \text{ W}$, respectively.

- Energy released from mitochondria as “wasted” energy in the course of citric acid cycle may be the most significant energy source for excitations of MT vibrations.

Whole metabolic turnover of *Saccharomyces cerevisiae* cell is 0.1 pW (10^{-13} W) (Lamprecht, 1980). Mitochondria are considered “power plants of the cell” generating ATP by means of energy from pyruvate and fatty acid. Less than 50% of energy is stored in ATP. The rest of the energy (“wasted” energy), i.e., more than 50%, is released to surroundings of mitochondria in the form of heat and emitted radiation (Fig. 4). Corresponding wasted power may be 0.05 pW . Since mitochondria are located near microtubules, part of the “wasted” power (we may assume 20%, i.e., 10^{-14} W) may be involved in excitation of the vibrations in MTs.

2. Method and Parameters Used for Calculation of Electric Field Around MT

First assessment of oscillating electric field around MTs has been done by Pokorný et al. (1997). Intensity of the electric field was evaluated along the lines parallel with the MT axis up to the distance of 500 nm. Their model included microtubules with heterodimers with random or coherent oscillating dipoles. Frequency of coherent excitations has been chosen as 100 MHz.

We made a preliminary assessment of MT electric field with different calculation method in Havelka and Cifra (2009a), Havelka (2008), Havelka and Cifra (2009b), but we used a higher value of dipole moment than corresponds to the real state. The complete

² $1.7 \text{ kcal/mol} = 7.1 \text{ kJ/mol} = 118 \times 10^{-22} \text{ J} = 0.074 \text{ eV}$.

³ $58.2 \text{ kJ/mol} = 0.97562 \times 10^{-19} \text{ J} = 0.609 \text{ eV}$.

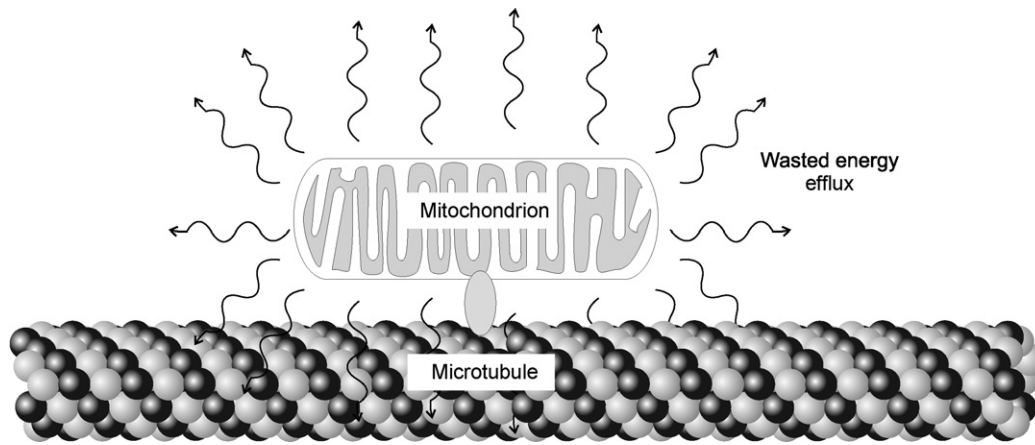


Fig. 4. Microtubule bathed in energy efflux from a mitochondrion. The mitochondrion not in scale to microtubule.

and more realistic results are presented in this paper. We analyze electric field generated by MT longitudinal axial vibration modes. We also study the spatial distribution of electric field of MT since this has not been performed before.

Due to axial longitudinal vibrations in the MT lattice an electric dipole moment will have an oscillating component in the direction parallel to MT axis. Oscillating dipole with the dimensions much smaller than wavelength of generated electric field can be represented by elementary oscillating electric dipole (Hertz dipole).

2.1. Components of the Electric and the Magnetic Fields of the Elementary Oscillating Electric Dipole

We use standard relations for the field of oscillating electric dipole in the form as provided in Novotný (1997):

$$\vec{H}_\phi = -\frac{Id}{4\pi} k^2 \sin \theta \left(\frac{1}{jkr} + \frac{1}{(jkr)^2} \right) e^{-jkr} \vec{\phi}_0 \quad (1)$$

$$\vec{E}_r = -\frac{Id}{4\pi} Z k^2 2 \cos \theta \left(\frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right) e^{-jkr} \vec{r}_0 \quad (2)$$

$$\vec{E}_\theta = -\frac{Id}{4\pi} Z k^2 \sin \theta \left(\frac{1}{jkr} + \frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right) e^{-jkr} \vec{\theta}_0 \quad (3)$$

H is magnetic field intensity; E is electric field intensity; I is equivalent current; d is length of the dipole; k is propagation constant; Z is wave impedance; j is imaginary unit and coordinate symbols are according to Fig. 5. None of these equations is dependent on ϕ and

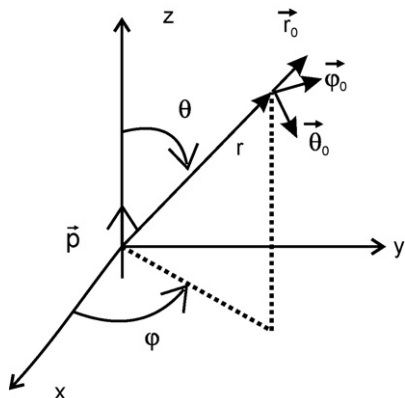


Fig. 5. A dipole in the Cartesian and spherical coordinate system.

all equations include:

$$Id = j\omega p \quad (4)$$

ω is angular frequency, and p is dipole moment ($Id = (\partial Q / \partial t)d = j\omega Qd$, $Qd = p$, Q is charge).

2.2. Description of the Electric Field Around a Microtubule

Electric field around MT can be calculated as a vectorial sum of elementary electric components generated by individual tubulin heterodimer dipoles. Frequency of the oscillations of individual dipoles has been set to the 1 GHz which falls into the spectrum predicted by Sirenko et al. (1996), Wang et al. (2006) and Wang and Zhang (2008) and also fits almost exactly the frequency of optical branches of vibration modes of microtubule calculated by Pokorný and Wu (1998) and Pokorný et al. (1997).

The near field around electric dipole has the electric component much stronger than the magnetic one. We concentrated therefore on the electric field component.

We have rewritten the Eqs. (2) and (3) for electric field in Cartesian coordinates which can be used for arbitrarily placed and tilted dipole in lossy medium. The procedure is described thoroughly in Cifra (2009). All calculations and figures have been made in MatLab.

2.3. Shift of Protofilaments

Although the MTs are tube-like structures, their wall does not have cylindrical symmetry. This is due to longitudinal shift of protofilaments in MT lattice. Accordingly, lattice with longitudinal shift of PTs simulates more realistic organization of protofilaments in the MT. This gives us more accurate information about the spatial distribution of electric field intensity of the around MT. We have chosen the configuration in A-lattice (Song and Mandelkow, 1993, 1995; Kikkawa et al., 1994; Amos, 1995), which is a lattice with a longitudinal shift of protofilaments of 4.92 nm (Table 1 and Fig. 6).

2.4. Types of Excitation and Calculation Parameters

Geometry and orientation of tubulin dipole moments in MT is approximately known. However, the elasto-electrical waves in

Table 1
Longitudinal shift of PT in A-lattice.

Number of PT	2	3	4	5	6	7
Shift to 1st PT (nm)	-4.92	-1.84	-6.76	-3.68	-0.6	2.48
Number of PT	8	9	10	11	12	13
Shift to 1st PT (nm)	-2.44	0.64	-4.28	-1.2	-6.12	-3.04

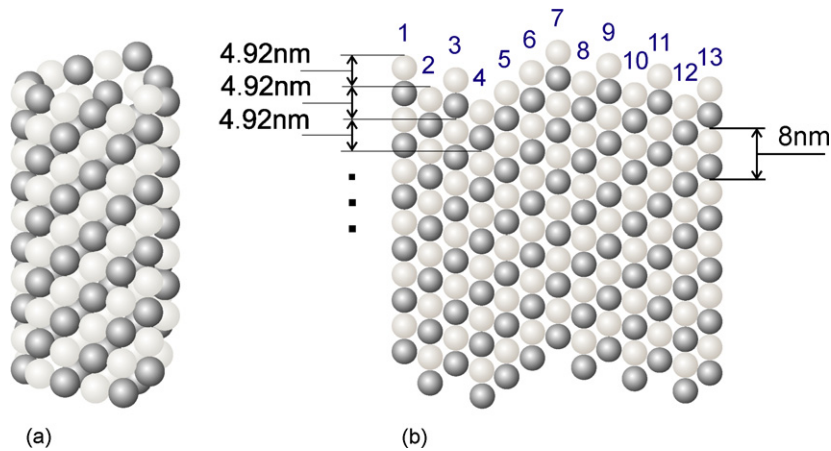


Fig. 6. (a) Microtubule with longitudinal shift of PT of 4.92 nm. (b) Shell of MT with longitudinal shift of protofilament.

microtubules also determine the efficiency, spatial distribution and dynamics of generated electric oscillations. There may be acoustical and optical modes of axial longitudinal oscillations. In optical modes, neighbor particles move in opposite direction. In acoustical modes, neighbor particles move in the same direction (except the end of Brillouin zone where the k -vector is large).

In the case of optical mode a component of dipole moment which is oriented along the axis would oscillate and generate electric oscillations.

Vibration modes which displace whole dipole along its axis would be significantly less efficient in generation of electric oscillations. This is due to the fact that opposite charges near each other and oscillating along the same axis with the same phase and frequency will behave as antiphase dipoles which effectively cancel each other field.

As shown in [Tuszyński et al. \(2002\)](#) and [Mershin et al. \(2004\)](#) a significant part of the tubulin dipole moment is oriented radially from MT axis. Certain kind of vibration modes involve radial oscillation of MT wall. Such kind of modes would involve displacement of rather large mass of surrounding cytosol, hence they would undergo stronger damping compared to the longitudinal optical oscillations in the direction along MT axis.

Circumferential elastic vibrations may be also connected with generation of oscillating electric field due to oscillatory changes of the distance between electric charges. Frequency of the modes depends on Young's and shear modulus. As the moduli exhibit large differences in magnitudes in different directions the frequencies of oscillations are corresponding.

Based on the discussion above, we have focused our analysis on the longitudinal optical oscillations of MT. Frequency of oscillations was chosen to be 1 GHz. It corresponds to calculations of MT vibrations of [Pokorný and Wang](#). We estimate the oscillatory displacement of the charge in tubulin dipole moment along the MT axis to be around 0.1 nm for the vibrations in GHz region.⁴ The static dipole moment (p_s) in the direction of the MT axis is accepted to be about 330 D ([Tuszyński et al., 2002](#); [Mershin et al., 2004](#)). Length of the tubulin heterodimer is 8 nm, so the effective oscillatory component of the dipole p_a at the frequency of 1 GHz may be estimated as $p_a = (0.1/8)p_s = 4.125 \text{ D} = 1.376 \times 10^{-29} \text{ C m}$.

⁴ This estimate is based on measurements of [Pelling et al. \(2004\)](#) and [Pelling et al. \(2005\)](#) who found oscillations of yeast cell membrane (likely excited by cytoskeleton) in the frequency range about 1 kHz and amplitudes about 1–10 nm. Microtubules associated with motor proteins are also able to vibrate at frequencies of hundreds of kHz and amplitudes up to 10 nm in cilia and flagella ([Kamimura and Kamiya, 1989, 1992](#); [Sakakibara et al., 2004](#)).

Excitation of vibrations in our calculations was based on following models:

- Random excitation—random phase of oscillations was approximated by modulation of dipole moment of each of dipoles by function $\sin(2\pi n)$, n = random number from interval (0, 1).
- Coherent excitation – modes with wavenumber k (k -modes) – longitudinal waves on the MT have been approximated by modulation of dipole moment of the individual heterodimers along the MT axis by function $a + b \times \sin(k_0 n z)$, z -axis corresponds to MT axis. Physically, these vibration modes belong to optical branch where neighbor particles (here α - and β -tubulin monomers in protofilaments) oscillate against each other. This corresponds to our approximation of heterodimer as an oscillating dipole. Number of mode n is provided instead of wavenumber k to indicate number of maxima. Note that direction of electric field vector is opposite in neighboring maxima as given by $\sin(k_0 n z)$ modulating function. In a special case all the dipoles may oscillate with the same phase (corresponds to $k = 0$ optical branch of longitudinal lattice vibration).

Dielectric parameters of the surrounding medium have been chosen based on experimentally obtained results from [Asami et al. \(1976\)](#) and [Asami and Yonezawa \(1996\)](#). Compared to the cited literature, we use slightly different value of the cytoplasm relative permittivity; instead 50 ([Asami et al., 1976](#)) we use 60. Slightly higher permittivity used corresponds physically to higher water content assumed ([Table 2](#)).

3. Results

Colormap figures depict MT electric field spatial distribution in the plane which contains the MT axis. White lines in the figures are contours of the MT wall. The arrows in the colormap figures denote the direction of the electric field vector. Note that neither arrow density nor arrow length is related to intensity. For greater dynamic range common logarithm of electric intensity is used for all

Table 2
Fixed parameters of calculations.

Frequency of vibrations	1 GHz
Number of protofilaments in MT	13
Number of heterodimers in each protofilament	50
Oscillatory component of dipole moment	$4.125 \text{ D} = 1.376 \times 10^{-29} \text{ C m}$
Relative permittivity of the medium	60
Conductivity of the medium	1 S/m

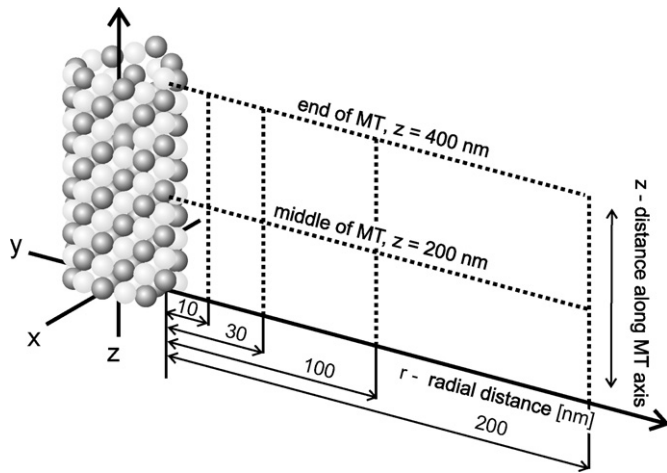


Fig. 7. The lines along which the electric field intensity has been plotted.

graphs and figures, e.g. color on the colorbar denoted 4 corresponds to intensity 10^4 V/m.

Electric field intensity dependence on distance radially from MT axis and along MT axis in selected distances from MT axis has been plotted in separate figures for simpler comparison. See Fig. 7 for the clarification of the plotting geometry.

3.1. Coherent Excitation

First we studied the simple model of the special case of coherent excitation. The spatial distribution of electric field around MT in this case is depicted in Fig. 8.

Coherently excited (at $k = 0$) MT behaves like a giant dipole having maxima of electric field intensity at its ends. This is of no wonder since it consists of chains of in phase oscillating dipoles with the same orientation of dipole moments.

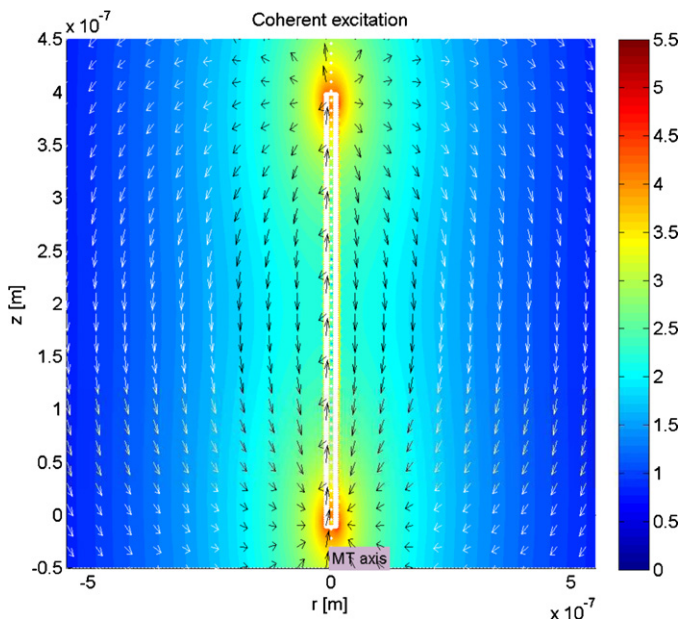


Fig. 8. Electric field around microtubule with coherent excitation ($k = 0$). Color coded electric field magnitude is in log scale ($\log(V/m)$). Arrows represent electric field vector.

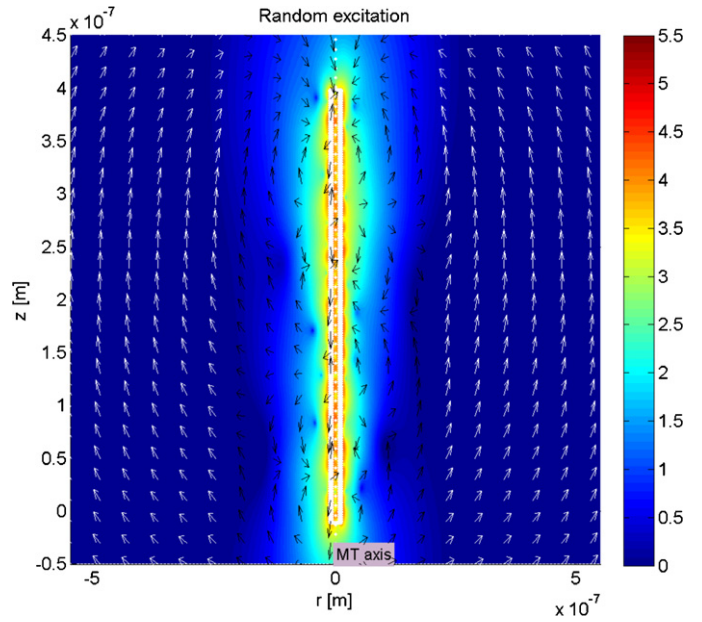


Fig. 9. Electric field around microtubule with random excitation. Color coded electric field magnitude is in log scale ($\log(V/m)$). Arrows represent electric field vector.

3.2. Random Excitation

Spatial distribution of electric field generated by random (thermal) excitation of MT is given in Fig. 9.

The spatial distribution depicted in the Fig. 9 is just one example which results from the certain random number sequence. Naturally, the spatial distribution of random excitation can take various forms and often exhibits local minima.

3.3. Comparison of Coherent and Random Excitation

Comparison of coherent ($k = 0$) and random excitation of electric field is provided in Fig. 10.

We clearly see the difference between the field generated by random and in phase (coherent) oscillations. Although electric field intensity is higher in the middle of the MT axis ($z = 200$ nm) at the surface of MT for random excitation, it falls off faster with the dis-

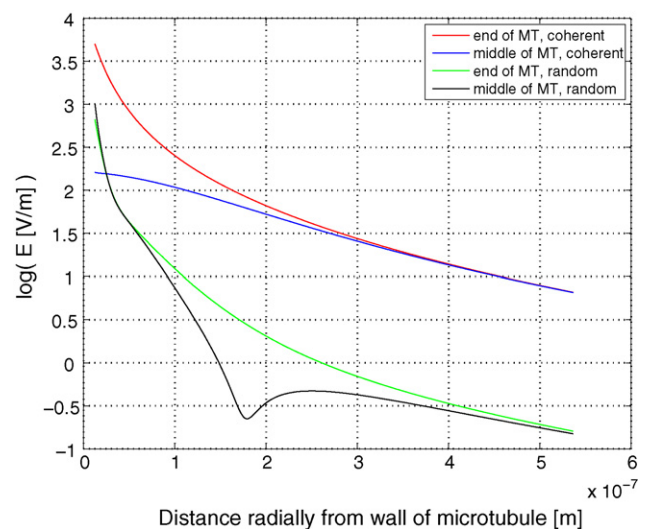


Fig. 10. Electric field intensity radial dependence from the wall of microtubule for the coherent and random excitation. From the end of MT ($z = 400$ nm) and from the middle of MT ($z = 200$ nm).

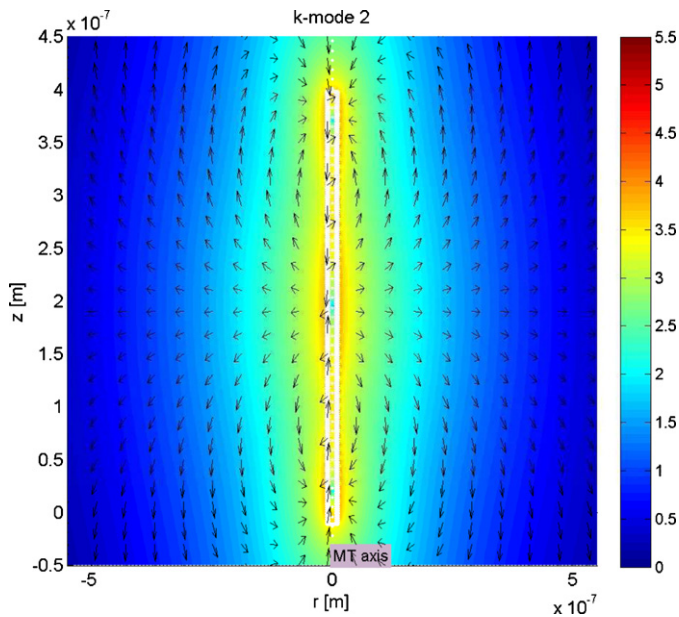


Fig. 11. Electric field distribution around MT of k -mode 2. Color coded electric field magnitude is in log scale ($\log(V/m)$). Arrows represent electric field vector.

tance compared to the in phase (coherent) excitation (Fig. 10). The electric field intensity is higher at the ends of MT ($z = 400$ nm and 0 nm⁵) for the in phase excitation compared to the random one (Fig. 10). There is a difference by one and a half orders of magnitude between the electric field intensity of randomly and in phase excited vibrations in the distance larger than ca. 100 nm.

3.4. Coherent Excitation of Higher k -modes

Normal vibration modes (termed as k -modes) can be excited in microtubules. We primarily chose to study k -modes number $n = 2, 5$ and 50 as the representative ones. We depict the spatial distribution of the k -mode 2 in Fig. 11 and zoomed in spatial distribution in 12.

Two maxima in the MT lattice are clearly discernible in Fig. 12 whereas in Fig. 11 they are not. Therefore we chose to provide only zoomed in figures when displaying spatial distribution of electric field for higher ($n > 0$) modes. Note that abscissa and ordinate have different scales in zoomed in figures; the r -axis (ordinate) is 10-fold stretched to resolve the details of electric field in direction radially from MT axis.

The zoomed in spatial distribution of electric field around MT of k -modes 5 and 50 is depicted in Figs. 13 and 14, respectively. Electric field of higher modes is depicted in figures in supplementary material.

We observed that every half-wave of k -mode of longitudinal lattice vibration produces effectively a k -mode “dipole” field⁶ the maxima of which are located on the poles of the “dipole”. Moreover, k -mode “dipoles” are oriented to each other with the same pole. This can be seen also from the orientation of electric field vectors (see, e.g. Fig. 12 where two k -mode “dipolar” fields are discernible in each intersection of the MT wall; the plus poles are in the middle of the MT wall, i.e. the electric field vector is oriented radially from the plus poles of the “dipole”). The effective number of maxima in

⁵ Values are not depicted for $z = 0$ nm but they are essentially the same as for $z = 400$ nm due to the symmetry.

⁶ We use term “ k -mode ‘dipole’” to describe an effective “dipole” field structure which appears in the MT wall due to every half wave of MT vibration. Number of mode indicates the number of k -mode “dipoles”.

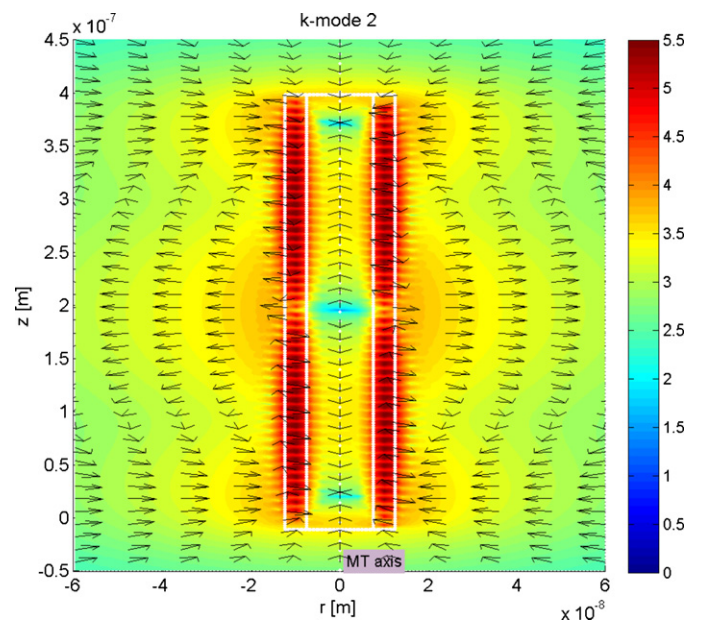


Fig. 12. Electric field distribution around MT of k -mode 2, zoomed in radial axis. Color coded electric field magnitude is in log scale ($\log(V/m)$). Arrows represent electric field vector.

the field intensity up to the distance of 10 nm from the MT wall (or more, depending on the mode number) for the k -modes $n > 0$ is actually $n + 1$ since each pole of k -mode “dipole” contributes by one maximum. The number of maxima further changes in greater distance.

Higher the mode, the total dipole moment of whole MT is split into many small k -mode “dipoles” with opposite directions. The MT obtains multipolar k -mode character with rising mode number. Multipolar character of field source distribution explains the steeper decrease of intensity from the MT for higher mode numbers (e.g. see Fig. 15).

We provide figures depicting electric field intensity radial dependence from MT end and middle of MT for k -mode 2, 5, 10, 20 and 50 in Figs. 15 and 16.

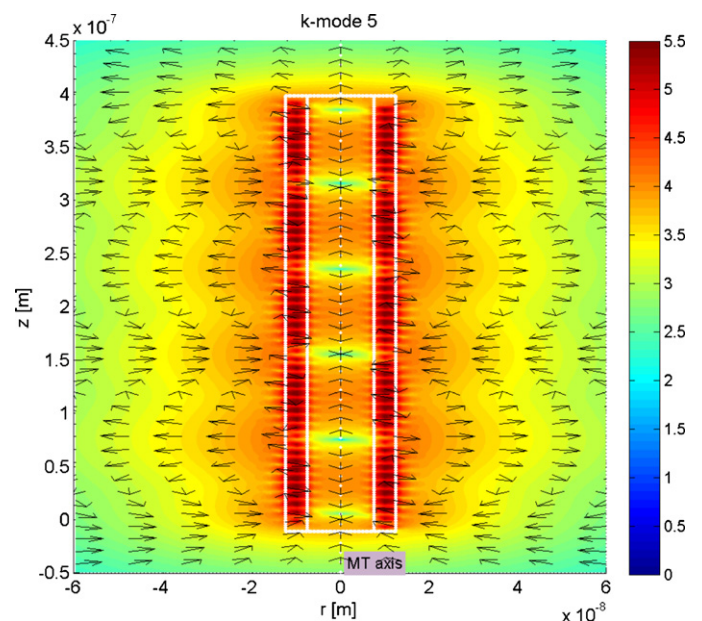


Fig. 13. Electric field distribution around MT of k -mode 5, zoomed in radial axis. Color coded electric field magnitude is in log scale ($\log(V/m)$). Arrows represent electric field vector.

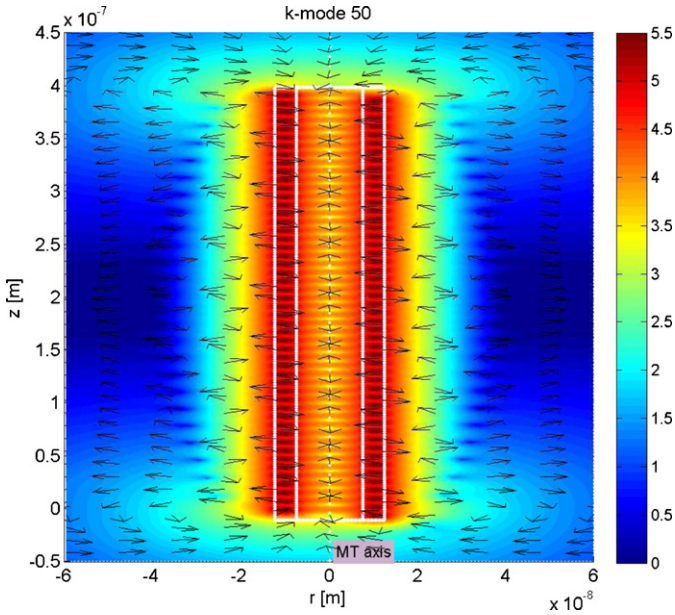


Fig. 14. Electric field distribution around MT of mode 50, zoomed in radial axis. Color coded electric field magnitude is in log scale ($\log(V/m)$). Arrows represent electric field vector.

At the close distance from the MT, for certain regions, there is a specific k -mode which has highest intensity, see Figs. 15 and 16. We see that different k -modes manifest different rate of decrease of intensity with increasing distance radially from MT wall. This resembles multipolar behavior of intensity dependence on distance: higher the multipole degree (k -mode number of MT vibrations) steeper the decrease of intensity from the multipolar source.

Following this line of thought, it is interesting to observe the profile of the intensity for various distances r from the microtubule. In Figs. 17, 18 and 19 we provide the profiles of intensity parallel to MT axis for k -mode 2, 5 and 50, respectively.

4. Discussion

First the frequency dependence will be discussed. For the frequencies up to high THz region, we can work with the microtubule generated field as near field in the distances we treat here (up

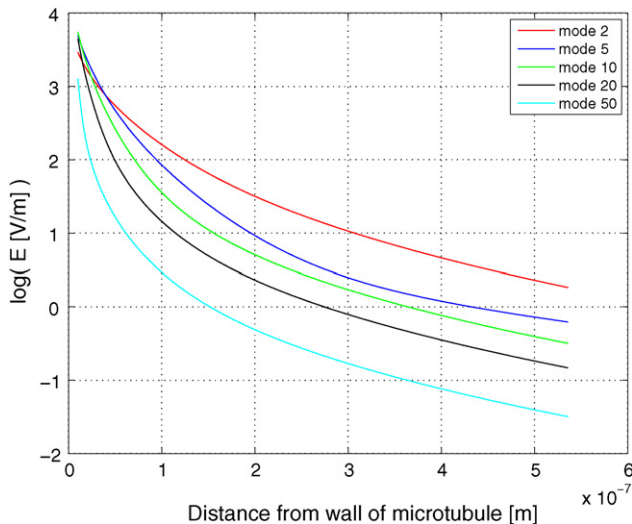


Fig. 15. Electric field intensity radial dependence from the wall of microtubule for the k -modes 2, 5, 10, 20 and 50. From the end of MT ($z = 400$ nm).

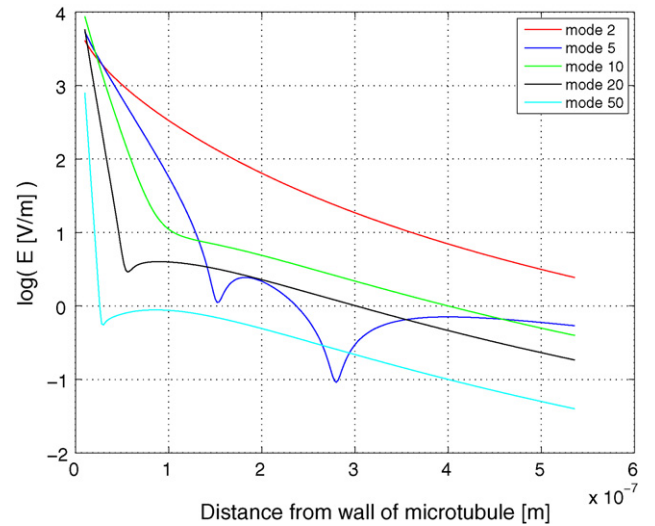


Fig. 16. Electric field intensity radial dependence from the wall of microtubule for the k -modes 2, 5, 10, 20 and 50. From the middle of MT ($z = 200$ nm).

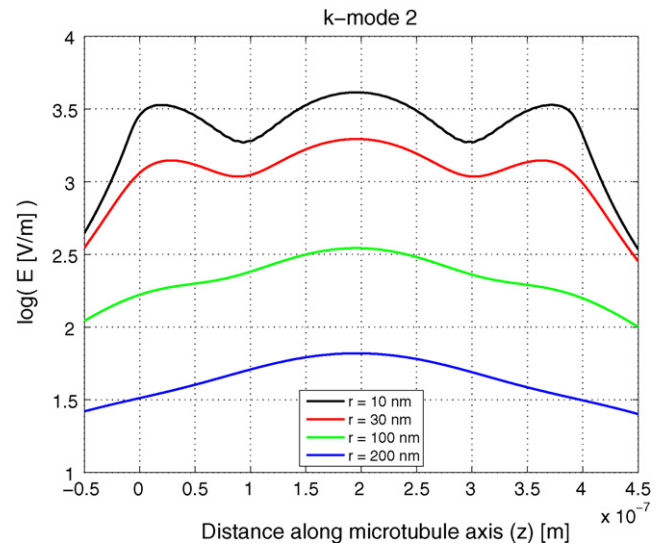


Fig. 17. Electric field intensity along the lines parallel to microtubule axis at the distance r from microtubule wall, mode 2. Microtubule is located from $z = 0$ to 400 nm.

to $1 \mu\text{m}$ from MT). So there is basically no difference in the field geometry for various frequencies.⁷ With changing frequency the magnitude of the intensity may change due to permittivity and conductivity relaxations. Increasing frequency has an effect on effective current created by dipole oscillations ($I_d = \omega p$), so the generated intensity would be higher if the amplitude of the vibrations does not decrease.

Discussing dipole moment separately; the generated intensity increases linearly with increasing dipole moment p magnitude. Amount of power in MT vibrations will influence effective value of dipole moment oscillating component p_a . Effective oscillating dipole moment would be increased if more power is supplied into the vibrations. If the amount of the oscillating charge is fixed,

⁷ Actually, there would be frequency dependence of the field geometry since wavenumber of vibration modes is bound to frequency of oscillations by dispersion relations. However, in the case of optical branch of longitudinal oscillations the dispersion relation is rather flat, i.e. the function of frequency on wavenumber is almost constant.

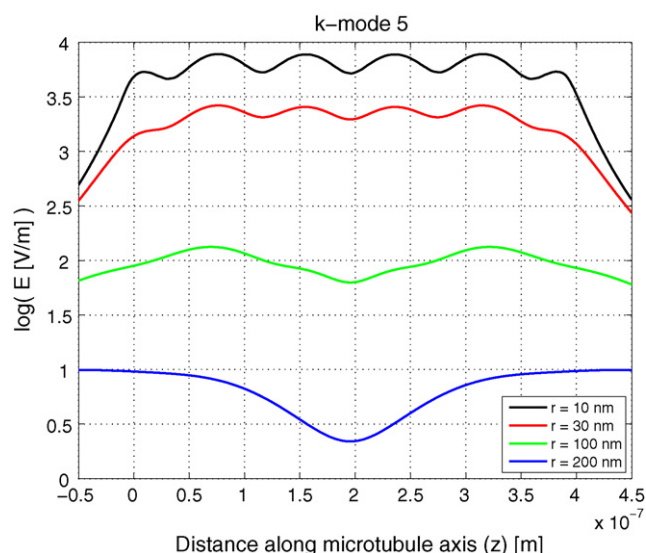


Fig. 18. Electric field intensity along the lines parallel to microtubule axis at the distance r radially from microtubule axis, mode 5. Microtubule is located from $z = 0$ to 400 nm.

increased dipole moment would correspond to the greater oscillating distance. However, for different frequencies, there is different amount of charge available for oscillations, given by the mobility of charge.

In a special case of strong energy condensation in certain microtubule vibration modes, electric field generated at certain frequency could be higher by several orders of magnitude and could exert force at the correspondingly greater distance. This kind of phenomenon may be transient only. Nevertheless, it may be involved in cellular interactions, if not by force interaction then by information transfer.

Only optical branches of vibrations can contribute significantly to electric field generation since in these kind of modes neighbor particles (monomers of tubulin) oscillate against each other. Actually, transverse (Kittel, 1976) and longitudinal (Berreman, 1963) optical modes in ionic crystals are responsible for electromagnetic field absorption. We can see the reciprocity of electromagnetic field

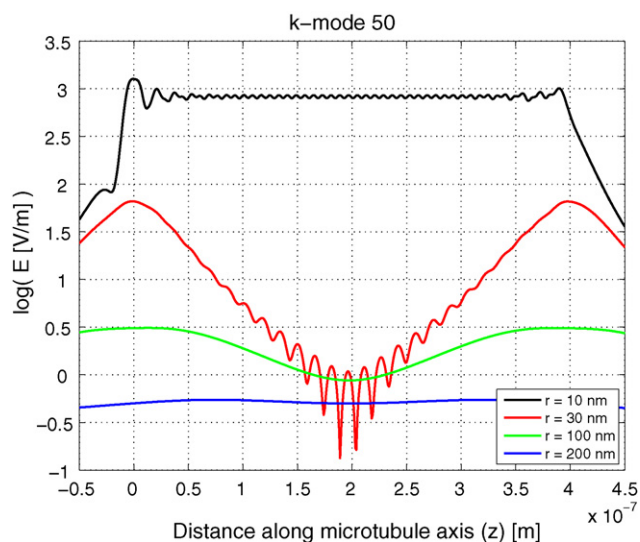


Fig. 19. Electric field intensity along the lines parallel to microtubule axis at the distance r from microtubule, mode 50. Microtubule is located from $z = 0$ to 400 nm. The slight asymmetry seen at the $r = 10$ nm is due to protofilament shift.

absorption and generation in these processes. Acoustic longitudinal branch could also effectively contribute to the electric field generation in the MT near the limit of the Brillouin zone, where the k vector is already large (vibration wavelength short) so that particles in the lattice oscillate against each other.

Microtubule associated proteins (MAPs) may also influence MT vibrations acting like a nodes imposing minimum of vibration amplitude (adding the mass to the binding point). This will influence generated field geometry, frequency of vibrations and the transfer of vibrations to other structures bound to the other end of MAPs.

In case of excitation of higher vibration modes, there will be regions of electric field with spatially oscillating direction of electric field vector. The dense distribution of regions with opposite field vector direction renders a problem for direct measurement of electric oscillations. Point detector needs to be used to resolve spatial regions of opposite direction of electric field vector.

We also studied the impact of the MT model with real lattice (protofilaments shifted to each other), which corresponds to realistic model of MT lattice compared to nonshifted models of MT (see [supplementary material](#)). For the low mode numbers ($n < N/2$), where N is the number of tubulin heterodimers in the MT, there was a small difference in the electric field around MT between the model with protofilaments (lattice) shifted and not shifted. However, the differences become significant for the growing mode number. The break in the symmetry introduced by protofilament shift changes the field geometry. The difference between the models with shifted and not shifted lattice are most pronounced for the higher mode number ($n > N/2$).

Vibrations of microtubules or of other electrically polar cellular structures may be the underlying mechanism generating the electric oscillations which were observed around living cells in kHz to MHz region (Pohl et al., 1981; Phillips et al., 1987; Hölzel, 2001; Pokorný et al., 2001; Jelínek et al., 2009).

We calculated and compared behavior of the electric field generated by random and coherent oscillations of microtubules. While random oscillations occur naturally without any special assumptions, the process where coherence of oscillations arise needs to be touched. Fröhlich's theory (Fröhlich, 1968a,b, 1980) describes a system of oscillators with energy supply, linear and nonlinear coupling with heat bath. If sufficient energy supply is provided to this system, condensation of energy occurs in lowest mode leading to its coherent excitation. Thus, coherent oscillations in microtubules described in this paper can be based on Fröhlich's theory. Numerous works have been inspired by Fröhlich's theory, for review see, e.g. Pokorný and Wu (1998), Hyland and Rowlands (2006) and Cifra (2009). Recently, Reimers et al. (2009) and McKemmish et al. (2009) presented analysis of Penrose–Hameroff orchestrated objective reduction (Orch OR) model of brain activity (Penrose and Hameroff, 1995; Hameroff, 1998; Hagan et al., 2002) with conclusion, that the model is founded on incorrect assumptions, in particular on Fröhlich's coherence, the essential requirement for Orch OR function. McKemmish et al. (2009) claimed that Fröhlich's coherence time is not long enough and that biochemical energy is too small to facilitate coherent condensation which follows from analysis in McKemmish et al. (2009). However, as described closer in comment by Pokorný (submitted for publication), authors of (McKemmish et al., 2009; Reimers et al., 2009) did not take into account the main sources of energy in the cell (“wasted” energy from mitochondria – see Section 1.2 this paper) and the fact that coherence may be strongly raised by

1. Weak damping due to organization of water (Fuchs et al., 2007, 2008, 2009; Giuliani et al., 2009) in strong static electric field around mitochondria (Tyner et al., 2007).

2. Interactions between individual Fröhlich's systems (Pokorný, submitted for publication).
3. Higher order nonlinear terms (Pokorný, 1982; Pokorný et al., 1986a,b).

If Fröhlich's theory of coherent excitations is reconsidered in the light of the facts in comment of Pokorný (submitted for publication), the disclaim of the Orch OR proposal of the brain function by Reimers et al. (2009) on the basis of insufficient coherence (McKemmish et al., 2009) does not seem to be justified.

5. Conclusions

Based on previous works (Pokorný and Wu, 1998; Pokorný, 1999; Pokorný et al., 1997, 1998; Havelka and Cifra, 2009a; Havelka, 2008), we have analyzed electric oscillations excited by vibrations of microtubule. Our model corresponds to optical branches of longitudinal vibrations.

The coherent vibrations of MT generate greater electric field in greater distance than random vibrations.

It was also found that the lattice shift of protofilaments has increasingly significant influence on the field geometry for the lattice vibration modes which wavelength is shorter than two tubulin heterodimer dipoles. This suggests necessity to model higher vibrations modes taking into account lattice shift, which is the real case.

More realistic oscillatory component of dipole moment based on experimental values of static dipole moment and amplitude of mechanical vibrations has been estimated than in previous works (Pokorný, 1999; Havelka and Cifra, 2009a).⁸ Using more realistic dipole moments electric field intensity around microtubules is about two orders of magnitude lower than calculated in previous works (Pokorný and Wu, 1998; Pokorný, 1999; Pokorný et al., 1997, 1998; Havelka and Cifra, 2009a; Havelka, 2008). Nevertheless, the electric field generated by microtubules may contribute to self-organization of biological processes also under these conditions.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.biosystems.2010.02.007.

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⁸ Previously, Pokorný has estimated oscillatory tubulin dipole moment at 100 MHz as $p = e \cdot d$, where e is elementary charge and d length of tubulin heterodimer (8 nm). It yields the value of $p = 1.48 \cdot 10^{-27}$ C m $\approx$ 500 D, the value of the static dipole moment. In our previous work, we have only taken the value of the total static dipole moment $\sim$ 1000 D.

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